Pre-Week Homework 05

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2.1

1. Every real number is an even integer. A statement and True
2. Every even integer is a real number. A statement and True
4. *Sets* $\mathbb{Z}$ *and* $\mathbb{N}$ Not a statement

2.5

1. $P \vee (Q \Rightarrow R)$

P	Q	R	$Q \rightarrow R$	$P \vee (Q \rightarrow R)$
T	T	T	T	T
T	T	F	F	T
T	F	T	T	T
T	F	F	T	T
F	T	T	T	T
F	T	F	F	F
F	F	T	T	T
F	F	F	T	T

4. $\neg(P \vee Q) \vee (\neg P)$

P	Q	$P \vee Q$	$\neg(P \vee Q) \vee (\neg P)$
T	T	T	F
T	F	T	F
F	T	T	T
F	F	F	T

8. $P \vee (Q \wedge \neg R)$

P	Q	R	$Q \wedge \neg R$	$P \vee (Q \wedge \neg R)$
T	T	T	F	T
T	T	F	T	T
T	F	T	F	T
T	F	F	F	T
F	T	T	F	F
F	T	F	T	T
F	F	T	F	F
F	F	F	F	F

2.6

4. $\neg(P \vee Q) = (\neg P) \wedge (\neg Q)$

P	Q	$\neg P$	$\neg Q$	$\neg(P \vee Q)$	$(\neg P) \wedge (\neg Q)$
T	T	F	F	**F**	**F**
T	F	F	T	**F**	**F**
F	T	T	F	**F**	**F**
F	F	T	T	**T**	**T**

12. $\neg(P \Rightarrow Q)$ and $P \wedge \neg Q$
I think these are equivalent because the negation of $P \Rightarrow Q$ would only be true when P is true and Q is false, which is the same condition for $P \wedge \neg Q$ to be true.

P	Q	$\neg Q$	$\neg(P \Rightarrow Q)$	$P \wedge \neg Q$
T	T	F	**F**	**F**
T	F	T	**T**	**T**
F	T	F	**F**	**F**
F	F	T	**F**	**F**

2.7

2. $\forall x \in \mathbb{R}, \exists n \in \mathbb{N}, x^n \geq 0$
For all real numbers x, there exists a natural number n such that x^n is greater than or equal to 0.
This is true because all real numbers raised to any natural number $2n$ will be positive and greater or equal to 0.

8. $\forall\ n \in \mathbb{Z}, \exists X \subseteq \mathbb{N}, |X| = n$
This is false because the cardinality of a set cannot be negative, so there is no subset of $\mathbb{N}$ with a cardinality of -1 for example.

2.9

5. For every $\varepsilon > 0$, there exists a $\delta > 0$ such that if $|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon$.

$$\forall\ \varepsilon \in \mathbb{N}, \exists\ \delta \in \mathbb{N}, (|x - a|) < \delta \implies (|f(x) - f(a)|) < \varepsilon$$

9. If x is a rational number and $x \neq 0$, then $\tan(x)$ is not a rational number.

$$x \in \mathbb{Q} \implies \tan(x) \notin \mathbb{Q}$$

2.10

4. For every positive number ε, there is a positive number δ such that $|x-a| < \delta$ implies $|f(x) - f(a)| < \varepsilon$.

$$\begin{gathered}
\forall\ \varepsilon > 0, \exists\ \delta > 0, (|x - a|) < \delta \implies (|f(x) - f(a)|) < \varepsilon \\
\neg(\forall\ \varepsilon > 0, \exists\ \delta > 0, (|x - a|) < \delta \implies (|f(x) - f(a)|) < \varepsilon) \\
\exists\ \varepsilon > 0, \neg(\exists\ \delta > 0, (|x - a|) < \delta \implies (|f(x) - f(a)|) < \varepsilon) \\
\exists\ \varepsilon > 0, \forall\ \delta > 0, \neg((|x - a|) < \delta \implies (|f(x) - f(a)|) < \varepsilon) \\
\exists\ \varepsilon > 0, \forall\ \delta > 0, (|x - a|) < \delta \wedge \neg((|f(x) - f(a)|) < \varepsilon)
\end{gathered}$$

$\exists\ \varepsilon > 0, \forall\ \delta > 0, (|x - a|) < \delta \wedge (|f(x) - f(a)|) \geq \varepsilon$
There exists a positive number ε such that all positive number δ are greater than $|x - a|$ and $|f(x) - f(a)|$ is greater than or equal to ε

8. If x is a rational number and $x \neq 0$, then $\tan(x)$ is not a rational number.

$$\begin{gathered}
\exists\ x \in \mathbb{Q} \wedge x \neq 0, \tan(x) \notin \mathbb{Q} \\
\neg(\exists\ x \in \mathbb{Q} \wedge x \neq 0, \tan(x) \notin \mathbb{Q}) \\
\forall\ x \in \mathbb{Q} \vee x = 0, \neg(\tan(x) \notin \mathbb{Q})
\end{gathered}$$

$\forall\ x \in \mathbb{Q} \vee x = 0, \tan(x) \in \mathbb{Q}$
For all rational numbers x or x equals 0, $\tan(x)$ is a rational number.